

Gang Jiang

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RESEARCH INTERESTS

Large Language Model (LLM) in Building, (Urban) Building Energy Modeling, High-Resolution Calibration, Physics-Informed Machine Learning, Fault Detection & Diagnosis

EDUCATION

- **University of Utah** Salt Lake City, UT, USA
Ph.D. in Civil Engineering Aug. 2022 – Jun. 2026 (Expected)
- **Tianjin University** Tianjin, China
M.Sc. in Architecture & Building Sciences; With Honor Aug. 2019 – Jun. 2022
- **Yancheng Institute of Technology** Yancheng, Jiangsu, China
B.Sc. Heating, Ventilation, & Air Conditioning; With Honor Aug. 2015 – Jun. 2019

RESEARCH EXPERIENCE

- **University of Utah, School of Computing** Salt Lake City, UT
Research Assistant May. 2025 - Present
 - **US-NSF Projects:** (Primary Contributor) Large language model (LLM) testing and development: **Conducted large-scale evaluation of LLM performance, designed benchmarking protocols, and contributed to refining model architectures and training strategies** for improved accuracy, efficiency, and robustness.
- **University of Utah, Civil & Environmental Engineering** Salt Lake City, UT
Research Assistant Aug. 2022 - May. 2025
 - **US-NSF #2311685:** (Primary Contributor) Developed a **natural language-driven auto-building energy modeling platform (EPlus-LLM, and Prompt-ABEM)** by integrating a large language model with physics-based simulation.
 - **Global Change & Sustainability Center:** (Primary Contributor) Developed **high-fidelity building energy models** (by Resistance-Capacity mechanism and EnergyPlus software) and **efficient calibration framework** that integrates an **enhanced Bayesian inference method** with a **deep-learning-based high-resolution surrogate model**, facilitating the development of Digital Twins for real-time monitoring and fault detection.
 - **US-Utah-DOT #24-8342:** (Primary Contributor) Designed a **scalable Vision-AI tool** for automated road condition assessment.
 - **US-NSF #2318720:** (Contributor) Conducted **co-simulations of thermo-responsive desiccants** for building dehumidification and decarbonization using EnergyPlus, Modelica, and Ansys.
- **Tianjin University** Tianjin, China
Research Assistant Aug. 2019 - Jun. 2022
 - **China-NSFs:** Developed **fault detection & diagnosis** methods for building systems with **incomplete data, integrating Modelica simulations with machine learning** to enhance accuracy and robustness.
 - **Industry Projects:** Conducted research on building energy efficiency and management, focusing on **optimizing energy efficiency and sustainability strategies**.
- **Amazon Web Services (AWS)** Beijing, China
Intern Jun. 2021 - Dec. 2021
 - **Data Center Design:** Assisted in **planning and optimizing** data center infrastructure, including local generators, uninterruptible power supplies (UPS), power distribution, cooling systems, and network architecture to enhance **resilience and scalability**.
 - **Data Center Operation:** Contributed to improving **energy efficiency and fault detection** in data centers through **operational optimizations and system monitoring**.

JOURNAL PUBLICATIONS

- [8] [G. Jiang](#), J. Chen. Efficient Fine-Tuning of Large Language Models for Automated Building Energy Modeling in Complex Cases. *Automation in Construction*, Jul. 2025.
- [7] Z. Ma, [G. Jiang](#), J. Chen. A Neural Ordinary Differential Equations-Based Approach for Enhanced Building Energy Modeling on Small Datasets. *Building Simulation*, Apr. 2025.
- [6] Z. Ma, [G. Jiang](#), Y. Hu, J. Chen. A Review of Physics-Informed Machine Learning for Building Energy Modeling. *Applied Energy*, Mar. 2025.
- [5] [G. Jiang](#), Z. Ma, L. Zhang, J. Chen. Prompt Engineering to Inform Large Language Models in Automated Building Energy Modeling. *Energy*, Feb. 2025.
- [4] Z. Ma, [G. Jiang](#), J. Chen. Physics-Informed Ensemble Learning with Residual Modeling for Enhanced Building Energy Prediction. *Energy and Buildings*, Nov. 2024.
- [3] [G. Jiang](#), B. Kuang, Z. Shi, Z. Ma, J. Chen. Snow Plow Performance Measures in Non-RWIS Locations. *National Academies of Sciences, Engineering, and Medicine*, Aug. 2024.
- [2] [G. Jiang](#), Y. Chen, Z. Wang, K. Powell, B. Billings, J. Chen. A Deep Learning-Based Bayesian Framework for High-Resolution Calibration of Building Energy Models. *Energy and Buildings*, Nov. 2024.
- [1] [G. Jiang](#), Z. Ma, L. Zhang, J. Chen. EPlus-LLM: A Large Language Model-Based Computing Platform for Automated Building Energy Modeling. *Applied Energy*, Aug. 2024.

CONFERENCE PUBLICATIONS

- [3] [G. Jiang](#), J. Chen. Automated Building Energy Modeling Using Large Language Model. *COBEE*, Jul. 2025.
- [2] [G. Jiang](#), J. Chen. Fine-Tuning Large Language Models for Automated Building Simulation. *BAS*, Jul. 2025.
- [1] [G. Jiang](#), L. Zhang, J. Chen. EPlus-LLM: A Novel Automated Building Simulation Platform Using Natural Language Processing. *ASHRAE Annual Conference*, Jul. 2024.

TEACHING EXPERIENCE

- **University of Utah** Salt Lake City, UT
 - A Course in Green Building & Building Energy Modeling* *Aug. 2024 - Dec. 2024*
 - **Computer Lab Instructor** : Led a 3-week hands-on session on building energy modeling using EnergyPlus & OpenStudio, focusing on sustainable building design, retrofit, and energy efficiency.
 - **Teaching Assistant** : Assisted with computer lab sessions, supporting course logistics, preparing class materials, slides, and syllabus, and helping students with hands-on coding assignments.

CONFERENCE & WORKSHOP PRESENTATIONS

- **ASHRAE CIDCO**: Automating Building Energy Modeling from Natural Language. Feb. 2026. [Invited Oral Presentation](#)
- **ASHRAE Annual**: EPlus-LLM: A Novel Automated Building Simulation Platform Using Natural Language. Jun. 2024. [Oral Presentation](#)

- **GCSC Environment and Sustainability Research Symposium:** Improved Bayesian Calibration of Campus Building Energy Models. Feb. 2023. [Poster](#)

SERVICES

- **GitHub Maintainer:** I maintain the open-source GitHub repositories [EPlus-LLM](#) and [Prompt-ABEM](#), and actively respond to issues.
- **Independent Reviewer:** *Cell Reports Physical Science*; *Renewable & Sustainable Energy Reviews*
- **Associate Member:** ASHRAE

PROGRAMMING SKILLS

- **Programming Languages:** Python, PyTorch, JavaScript
- **Technologies:** High-Performance Computing on Linux, Fine-Tuning & Prompt Engineering for LLMs, Retriever-Augmented Generation, Physics-Informed Modeling, RC (Resistance-Capacity) Modeling, High-Resolution Bayesian Calibration, Incomplete Data-based Fault Detection & Diagnosis
- **Software:** EnergyPlus, SketchUp, Matlab, Dymola (Modelica)